

$\mu \rightarrow e\gamma$

General calculation

We consider the amplitude

$$\mathcal{M}(\mu \rightarrow e\gamma) = e \epsilon^{*\mu}(q) \langle e | J_\mu^{\text{em}} | \mu \rangle. \quad (1)$$

The most general Lorentz structure is

$$\langle e | J_\mu^{\text{em}} | \mu \rangle = \bar{u}_e(p-q) \left[i\sigma_{\mu\nu} q^\nu (A + B\gamma_5) + \gamma_\mu (C + D\gamma_5) + q_\mu (E + F\gamma_5) \right] u_\mu(p). \quad (2)$$

Using current conservation (Ward identity),

$$q^\mu J_\mu^{\text{em}} = 0, \quad (3)$$

we obtain

$$q^\mu \bar{u}_e \left[i\sigma_{\mu\nu} q^\nu (A + B\gamma_5) + \gamma_\mu (C + D\gamma_5) + q_\mu (E + F\gamma_5) \right] u_\mu = 0. \quad (4)$$

Since $q^\mu \sigma_{\mu\nu} q^\nu = 0$, this reduces to

$$\bar{u}_e \left[\not{q} (C + D\gamma_5) + q^2 (E + F\gamma_5) \right] u_\mu = 0. \quad (5)$$

Using Dirac equations,

$$\not{p} u_\mu = m_\mu u_\mu, \quad \bar{u}_e \not{p}' = m_e \bar{u}_e, \quad (6)$$

one finds

$$m_e (C + D\gamma_5) - m_\mu (C - D\gamma_5) = 0, \quad (7)$$

which implies

$$C = D = 0. \quad (8)$$

Thus the amplitude simplifies to

$$\mathcal{M} = e \epsilon^{*\mu} \bar{u}_e [i\sigma_{\mu\nu} q^\nu (A + B\gamma_5)] u_\mu. \quad (9)$$

Define

$$A_{R,L} = A \pm B. \quad (10)$$

Then

$$|\mathcal{M}|^2 = m_\mu^2 (|A_L|^2 + |A_R|^2), \quad (11)$$

after spin and polarization sum.

The decay width is

$$\Gamma(\mu \rightarrow e\gamma) = \frac{1}{16\pi} m_\mu^3 (|A_L|^2 + |A_R|^2). \quad (12)$$

SM calculation

In the SM (in R_ξ gauge), there is a single diagram with W and neutrinos in the loop.

The amplitude schematically is

$$\mathcal{M}^{(1)} \sim \int \frac{d^4 k}{(2\pi)^4} \bar{u}_e \gamma^\alpha P_L \frac{\not{p} - \not{k} + m_\nu}{(p-k)^2 - m_\nu^2} \gamma^\beta P_L u_\mu \frac{1}{k^2 - m_W^2} \frac{1}{(k+q)^2 - m_W^2} \Gamma_{\alpha\beta\mu}. \quad (13)$$

The three-boson vertex is

$$\Gamma_{\alpha\beta\mu}(k, q) = g_{\alpha\beta}(2k+q)_\mu - g_{\beta\mu}(k+2q)_\alpha - g_{\mu\alpha}(k-q)_\beta. \quad (14)$$

Expanding the propagator in m_ν^2 ,

$$\frac{1}{(p-k)^2 - m_\nu^2} \simeq \frac{1}{(p-k)^2} + \frac{m_\nu^2}{(p-k)^4} + \dots \quad (15)$$

Using unitarity of the PMNS matrix,

$$\sum_k U_{\mu k} U_{ek}^* = 0, \quad (16)$$

the leading term cancels, and only the mass-suppressed term survives:

$$\sum_k U_{\mu k} U_{ek}^* m_{\nu_k}^2. \quad (17)$$

Thus,

$$A^{(1)} \sim \frac{g^2 e}{128\pi^2} \frac{m_\nu^2}{m_W^2}. \quad (18)$$

Result

The branching ratio behaves as

$$\Gamma(\mu \rightarrow e\gamma) \propto \left| \sum_k U_{\mu k} U_{ek}^* \frac{m_{\nu_k}^2}{m_W^2} \right|^2 \sim 10^{-44}. \quad (19)$$

EFT calculation

(To be completed.)